

SLOPEWATCH · CALCULATION

SlopeWatch sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-01	Amish Chadha	Re-run for the constructable design (SLW-DDR-003); crack gauge rod, side-mounted alert unit, hole depth, install time and cost updated; cost reported against the value-engineering target
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (tilt resolution, temperature drift with pipe conduction, crack gauge, airtime and latency, alarm reach, energy and rail current, bus cable, mast wind load, installation, data, cost)

SlopeWatch sizing calculations

On paper, SlopeWatch meets six of its thirteen requirements (four by calculation, two by design), has four at risk, misses none, leaves two that only a field trial can show, and is over its value-engineering target on cost (R11). This issue (v0.3) re-runs the calculations for the constructable design of SLW-DDR-003: the parts added to make the design buildable bring the reference site to \$296.50, \$46.50 over the \$250 value-engineering target, and the side-mounted alert unit lowers the optional mast's factor on yield from 2.1 to 1.7. It keeps Amish's decisions of 2026-09-25 (SLW-DDR-002): the reference site mounts the node on an existing pole, with a new mast as a site option; and the keyed silence switch moves to its own post about 5 m from the siren, which cuts the level there from about 105 to about 95 dB(A). The four at risk are the remote alert time at the slowest radio setting (R6), siren reach (R7), the FieldNode 12 V rail current during an alarm (R8) and the FieldNode enclosure temperature (R9). The v0.1 calculations changed four details of the TRL 2 concept: the sensor capsule moves from 0.3 m to 0.4 m deep, because at 0.3 m wet soil lets the daily temperature cycle through faster than R2 allows; the crack gauge gets its own RS-485 reader; the precaution beacon is limited to 1 % duty, because a 5 % slow flash would drain the cell in a long precaution spell; and the crack-gauge precaution rate is taken over 24 h. Every number here is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [B5], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that any slope is safe or that an alarm will come in time. SlopeWatch cannot warn of brittle failures, deep-seated movement or failures between its stakes. See SLW-PRC-001, Safety.

■ Scope and method

The note checks every requirement in SLW-REQ-001 v0.5 against the design in SLW-PRC-001 v0.5, the constructable design of SLW-DDR-003 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `SITE` and `derived()`, so the pipe, capsule depth, grout column, gauge, mast and cable run used here are the ones in the STEP files and in drawing SLW-DWG-001. It reads `bom/bom.csv` and `budget_usd` in `project.yaml`, and writes the results table to `docs/04-calcs/results.csv`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

The design case is the reference site of SLW-REQ-001: three stakes 10 m apart on the fall line of a 25 degree face (20 m of face, 8.5 m of rise), a crack gauge 8 m above the top stake, and the node on an existing pole (or the optional mast) 15 m beyond the toe, with the keyed switch on its own post about 5 m from the siren, and ambient -10 to 50 °C at the surface.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Inclinometer	SCL3300-D01 in mode 1: ± 90 degrees, noise density 0.0024 degrees per $\sqrt{\text{Hz}}$, angle output 0.0055 degrees per LSB; offset temperature drift ± 0.005 degrees per K	Murata datasheet and product page , both checked 2026-09-25
Reading	Bus on for 12 s every 10 min: 2 s start-up and a 10 s average	Design choice
Soil temperature	Sinusoidal daily cycle; surface range 20 K in moist and wet soil, 30 K in dry soil; diffusivity 0.3, 0.6 and 1.0 mm^2/s and conductivity 0.4, 1.0 and 1.6 W/mK for dry, moist and wet soil	Handbook ranges for mineral soils; to be confirmed at a site
Pipe heating	160 mm of bare pipe between ground and head in full sun (900 W/m^2 , absorptance 0.6), $10 \text{ W/m}^2\text{K}$ to air; steel 50 W/mK ; buried pipe treated as a fin losing heat to the soil; steady state (no heat capacity), so an upper bound	Screening values
Crack gauge	12-bit converter on a 100 mm potentiometer of 0.1 % linearity; steel rod and aluminum body see a 30 K daily range under the guard	Typical parts
Radio	LoRaWAN EU868, 125 kHz, coding rate 4/5, 20-byte payload plus 13 bytes of overhead; 45 mA at 3.3 V while transmitting; network server 5 s, alert service 30 s and SMS delivery 60 s	Semtech airtime formula; service times assumed
Siren	110 dB(A) at 1 m; spherical spreading; air absorption 0.015 dB/m near 3 kHz; 0 to 3 dB excess ground attenuation	Typical 12 V piezo siren; order of magnitude from ISO 9613-1
Energy	FieldNode cell 3.2 V, 6 Ah, 80 % usable, 85 % of capacity at -10°C ; rail converters 90 %; core 4 mWh/day (FND-CAL-001); capsule 6 mA and reader 5.5 mA at 5 V; siren 0.25 A and beacon 0.2 A at 12 V	FieldNode figures; parts typical; cold factor assumed
Wind	35 m/s gust (735 Pa); drag coefficients 1.2 (panel, pole, alert unit) and 1.3 (boxes); medium soil 18 kN/m^3 , $K_p = 3$	Screening values as in FND-CAL-001 and WWT-CAL-001; not a code check

■ A. Tilt measurement (R1)

- **Resolution and range.** The angle output steps by 0.0055 degrees in every mode. Mode 1 spans ± 90 degrees; modes 3 and 4, the low-noise inclination modes, stop at ± 10 degrees [A1], which would not meet the ± 30 degree range of R1. The firmware must therefore use mode 1.
- **Noise.** A 10 s average in mode 1 has 0.00054 degrees of noise [A2]. A least-squares rate over 1 h (7 readings) has 0.00061 degrees per hour of noise, 164 times below the warning rate; over 3 h (19 readings) it is 74 times below the precaution rate [A3], [A4].
- **Meaning.** The warning rate of 0.1 degrees per hour is 1.75 mm per hour at 1 m from the pivot, and the precaution change over 3 h is 5.5 output steps [A5]. **R1 is met on paper.**

■ B. Temperature drift of a buried capsule (R2)

The daily soil temperature wave shrinks with depth by $\exp(-z/d)$, where d is the damping depth, 91 to 166 mm for dry to wet soil [B2]. Heat also runs down the steel pipe from its sunlit exposed length, which absorbs 4.17 W in full sun [B1] and would sit up to 8.8 K above the soil at ground level [B3]; the buried pipe sheds that heat to the soil within about 0.1 m.

Table 2. Daily temperature range at the capsule and apparent tilt, against R2 (0.02 degrees per day and 0.002 degrees per hour) [B2].

Capsule depth	Soil, surface range	Soil range at capsule	Added by pipe	Apparent tilt per day	Peak apparent rate	R2
0.3 m	Dry, 30 K	1.10 K	1.36 K	0.0123 degrees	0.00161 degrees/h	Met
0.3 m	Moist, 20 K	1.94 K	0.34 K	0.0114 degrees	0.00149 degrees/h	Met
0.3 m	Wet, 20 K	3.28 K	0.13 K	0.0170 degrees	0.00223 degrees/h	Not met
0.4 m	Dry, 30 K	0.37 K	0.82 K	0.0059 degrees	0.00078 degrees/h	Met
0.4 m	Moist, 20 K	0.89 K	0.15 K	0.0052 degrees	0.00068 degrees/h	Met
0.4 m	Wet, 20 K	1.79 K	0.05 K	0.0092 degrees	0.00120 degrees/h	Met

- **The TRL 2 depth misses R2 in wet soil.** Wet soil carries the daily wave deeper, and the peak apparent rate at 0.3 m exceeds 0.002 degrees per hour. At 0.4 m every case passes; the design case (wet soil) has a margin of 1.66 on the hourly limit and 2.18 on the daily limit, and the precaution threshold is 8.3 times the thermal rate [B5]. The capsule now sits 0.4 m deep, the depth R9 already names.
- **Pipe conduction matters in dry soil.** In dry soil the pipe adds more than the soil does (0.82 K against 0.37 K at 0.4 m), because dry soil conducts heat away poorly. The steel pipe still passes, so the TRL 2 question of steel or plastic pipe does not need a plastic pipe for R2. Closed-cell foam plugs above and below the capsule stop air in the pipe from carrying heat down.
- **Unburied, the sensor would fail.** At the surface the same sensor would show 0.100 degrees per day and 0.0131 degrees per hour [B4], above the precaution threshold every sunny day. The TRL 2 figures stand.
- **R2 is met on paper.** It rests on the handbook soil values and on a steady-state pipe model; rain soaking into warm ground can change soil temperature faster than a daily sine wave, which only site data can show.

■ C. Crack gauge (R3)

- **Resolution.** A 12-bit reading of the 100 mm stroke steps by 0.0244 mm; the 0.1 % linearity class gives 0.10 mm of absolute error [C1]. The rate of opening, which drives the alerts, depends on repeatability rather than linearity. **R3 is met by design** (datasheet class); re-setting after 100 mm needs only an Allen key on a clamp block set screw.
- **Thermal swing.** The 501 mm steel extension rod and 260 mm aluminum body under the guard change length by 0.36 mm over a 30 K day, a peak apparent rate of 0.047 mm per hour [C2]. Over a 3 h window that reads as up to 1.13 mm per day, above the 1 mm per day precaution placeholder; over a 24 h window it cancels [C3]. The crack-gauge precaution rate is therefore taken over 24 h. The 1 mm per hour warning placeholder is 21 times the thermal rate, and at that rate the stroke lasts 100 h [C4].

- **Reader.** The gauge sits about 60 m of cable from the node, too far for a clean analog signal on a cable shared with the RS-485 bus. It gets the capsule's board without the inclinometer, in a small IP67 box (BOM line 4, +\$6).

■ D. Sampling, airtime and alert latency (R4, R5, R6)

- **Airtime.** A 20-byte uplink takes 72 ms at SF7, 247 ms at SF9, 453 ms at SF10 and 1,810 ms at SF12 [D1]; the TRL 2 figures of 0.4 s and 1.5 s were low. Six uplinks an hour use 0.08 % of the time at SF10 and 0.30 % at SF12, within the 1 % EU868 limit [D2]. **R4 is met on paper.**
- **Fair use.** On a private TwinKit gateway only the duty cycle applies. On The Things Network, whose fair-use policy allows 30 s a day, hourly uplinks fit at SF9 and SF10 (5.9 s and 10.9 s) but not at SF12 (43.5 s), and 10 min uplinks in the precaution or warning state exceed it from SF10 upward (65.2 s at SF10) [D3]. An alert state is short, but a site that lives at SF12 on The Things Network exceeds fair use, so under SLW-DDR-002 such a site uses its own TwinKit gateway.
- **Local alarm.** From the start of the reading that confirms a warning to the siren takes 13.4 s: bus start-up, the 10 s average, polling four devices and the decision [D4]. **R5 is met by design** (60 s).
- **Remote alert.** Warning to SMS takes 95 to 97 s at the first try. If that uplink is lost, the duty-cycle wait before a retry is 45 s at SF10 and 179 s at SF12, giving 141 s (2.3 min) and 278 s (4.6 min) [D5]. **R6 is at risk:** met on paper where coverage exists, with little margin at SF12. A site without LoRaWAN coverage needs a gateway (TwinKit, about \$290 in parts) or gets no remote alert.

■ E. Alarm reach (R7)

- **Open ground.** The siren gives 65.5 to 68.5 dB(A) at 100 m [E1] and reaches 65 dB(A) out to about 105 to 139 m [E2]. **R7 is at risk:** met in open, quiet ground with 0.5 dB to spare over soft ground, before any allowance for a siren that falls short of its rating or for wind.
- **Near machinery.** Against 80 dB(A) of running plant, an alarm 15 dB above ambient reaches only about 5 m [E3]. Sites with machinery need a second alert unit at the work face or a radio pager for operators (D7 option).
- **Close range.** The siren now sits on the side-mounted alert box, its centre 3.23 m above ground on the mast (SLW-DDR-003; 3.26 m before). A switch box on the mast, as in v0.1, would sit at about 106 dB(A), and 2 m from the mast the level is about 102 dB(A); at 106 dB(A) the NIOSH limit allows about 4 min [E4]. Under SLW-DDR-002 the keyed switch stands on its own post about 5 m from the mast, on a 7 m lead, where the level is 95.5 dB(A) and the NIOSH allowance about 42 min [E5]. Silencing takes seconds; people should still not stand at the mast during an alarm.

■ F. Energy and rail current (R8)

- **Sensor load.** The bus draws 1.44 J per reading, 2.67 mW average from the cell, or 0.064 Wh per day [F1]. The TRL 2 figure of under 1 mW assumed 2 s of power per reading; the 10 s average raises it, still far below the FieldNode 100 mW sensor allowance.
- **Uplinks.** Hourly uplinks sit inside the FieldNode core budget, which assumes a 15 min report. Uplinks every 10 min at SF12 add 10.8 mWh per day [F2].
- **Alarm.** The siren and beacon take 5.4 W at 12 V, 6.0 W from the cell: 3.00 Wh for a 30 min alarm, plus 0.67 Wh for a 30 min silenced period with the beacon flashing at 50 % [F3].
- **Precaution beacon.** A slow flash at 5 % duty would cost 3.20 Wh per day; at 1 % it costs 0.64 Wh per day [F4]. Five sunless days in precaution with one alarm need 20.06 Wh at 5 % duty, more than the 13.06 Wh the cell

holds at -10 °C, but 7.26 Wh (56 %) at 1 % duty [F5], [F7]. The beacon's slow flash is therefore limited to 1 % duty (for example 100 ms every 10 s). In the normal state the same five days with one alarm need 4.01 Wh [F6]. The average load in precaution is 29.8 mW, within the FieldNode 100 mW allowance [F8].

- **Rail current.** The alarm draws 0.45 A from the 12 V rail and 2.00 A from the cell at 3.0 V (0.33 C, within the 5 A cell fuse) [F9]. FieldNode does not state the 12 V rail's current rating. **R8 is met on energy and at risk on the rail current** until FieldNode specifies at least 0.5 A continuous on its 12 V rail. Under SLW-DDR-002 that request goes to FieldNode (a cross-repo action), rather than giving the alert unit its own battery.

■ G. Sensor bus cable (R4, R9)

- **Length.** The reference layout needs 53.8 m of cable, 59.2 m with 10 % slack, within the 60 m in the BOM [G1].
- **Voltage drop.** The 0.5 mm² loop measures 4.07 Ω; with all 23.5 mA drawn at the far end it drops 0.096 V of the 5 V rail, leaving 1.30 V above the capsules' regulators' need [G2]. The bus uses the FieldNode 5 V port rather than 12 V, because the capsules' linear regulators would waste more than half the power at 12 V. RS-485 over 59 m is far inside its 1,200 m class limit [G3].
- **Ports.** The bus (5 V, ground, A, B) takes one FieldNode M12 5-pin port. The alert unit takes the other: 12 V, ground, siren drive, beacon drive and the silence switch, which fits FieldNode's one-rail-per-port rule (FND-DDR-001). The FieldNode pinout itself is still open (FND-DDR-001, O2).

■ H. Mast in wind

- **Loads.** The mast is a site option (SLW-DDR-002); where it is used, a 35 m/s gust puts 51 N on the panel, 29 N on the node enclosure, 81 N on the alert box, horn, beacon and the part of its back plate below the box, and 136 N on the mast: 298 N and a base moment of 653 N·m [H1]. The switch box is no longer on the mast. The side-mounted alert unit of SLW-DDR-003 adds its back plate to the wind area, which v0.2 (48 N, 541 N·m) did not count.
- **Mast.** The 48.3 x 3.2 mm pipe sees 136 MPa, a factor of 1.7 on the 235 MPa yield [H2], and deflects 79 mm at the top [H3]. The alert unit on the mast adds about 70 % to the base moment of a node alone. An existing pole must carry the same loads; the installer checks it by eye and with the site owner.
- **Footing.** The 320 mm by 600 mm footing resists 668 N sideways in medium soil, a factor of 2.2 [H4]. Soft or wet ground needs a site check. This is not a code check.

■ I. Installing one stake (R10)

- **Materials.** Each hole is 110 mm by 829 mm (7.9 L of spoil), deep enough for 20 mm of grout under the screwed end cap; the grout column takes 3.68 L, about 7.4 kg of dry mix [I1].
- **Time.** Augering (15 min), mixing (5 min), setting and grouting (5 min), backfilling (8 min), fitting the stand tube, capsule, foam plug, head and conduit fittings and joining the bus (8 min) and checking the reading (3 min) add up to 44 min against the 45 min of R10 [I2]. The grout then sets for 24 h before the baseline starts [I3]. **R10 is not verifiable at TRL 3:** the estimate is at the limit and the auger rate depends on the ground; only a timed installation can show it.

■ J. Data (R12)

A 28-byte record per reading gives 363 kB for 90 days in binary, or 1.04 MB as CSV, against the 16 MB FieldNode flash [J1]. **R12 is met on paper.**

■ K. Cost (R11)

`budget_usd` is a hypothetical value-engineering target, not a spending limit (STANDARDS section 18). The BOM has 10 lines, all priced, totaling \$450.50 with the FieldNode core and the optional mast [K1]. Under SLW-DDR-001 D1 the FieldNode core (\$126) is costed in FieldNode, and under SLW-DDR-002 the reference site mounts the node on an existing pole.

Value-engineering target: USD 250. Estimated cost of the constructable design: USD 296.50 for the reference site (USD 46.50, or 18.6 %, over the target) [K2]. A site that needs the mast and footing (line 8, \$28) comes to \$324.50, \$74.50 over the target; the reference site with the FieldNode core comes to \$422.50 [K3]. The three stake sets cost \$154.50, \$51.50 per extra stake plus about 10 m of cable [K4].

The constructable design (SLW-DDR-003) added \$51.50 to the reference site: screwed end caps and stand tubes (\$12 for three stakes), centring collars (\$1.50), stake heads from stock drainage fittings with two conduit fittings and lever connectors (\$18), the crack gauge clamp blocks, ball joints, rod coupling, folded guard and pegs (\$8), the alert unit's back plate, V-blocks and band clamps and the switch back plate and hose clips (\$10), and tape and a field-wireable plug (\$2). Earlier, the TRL 3 changes added \$6 for the crack-gauge reader and \$1 for foam plugs, and SLW-DDR-002 added \$8 for the switch post and lead. **R11 is over the value-engineering target by \$46.50.** The main cost drivers and the savings worth trying are listed in the design decisions register (SLW-DEC-001). A LoRaWAN gateway, where needed, is outside the site cost.

■ L. Results against every requirement

Table 3. Requirement status from this note [L]. At-risk items first; none is unmet; R11 is reported against the value-engineering target.

ID	Requirement	Value	Target	Status
R6	Remote alert	1.6 min at the first try; 4.6 min worst at SF12 with one lost uplink	5 min, where coverage exists	At risk
R7	Alarm audible	65.5 to 68.5 dB(A) at 100 m in open ground; about 5 m near running plant	65 dB(A) at 100 m	At risk
R8	Energy autonomy	7.3 Wh of 13.1 Wh in the worst case; 0.45 A on a 12 V rail with no stated rating	5 days, one 30 min alarm	At risk (rail current)
R9	Survive burial and weather	Capsule, head and gland IP67 by parts; FieldNode interior above 60 °C in 45 °C sun (FND-CAL-001)	IP67 at 0.4 m; node IP65; -10 to 50 °C	At risk
R1	Measure surface tilt	0.0055 degree step, 0.00054 degree noise, ±90 degrees in mode 1	0.01 degrees, ±30 degrees	Met on paper
R2	Limit false tilt from temperature	0.0092 degrees per day, 0.0012 degrees per hour (wet soil, 0.4 m)	0.02 per day, 0.002 per hour	Met on paper
R4	Sample and report	10 min readings; 10.9 s per hour at SF12	10 min; 60 and 10 min uplinks	Met on paper
R12	Open, local data	363 kB for 90 days	90 days, CSV, any server	Met on paper
R11	Affordable	\$296.50 for the reference site (existing pole); \$324.50 with the optional mast	\$250 value-engineering target, SlopeWatch-specific parts, reference site	Over the value-engineering target by \$46.50
R3	Measure crack opening	100 mm stroke, 0.024 mm step, 0.10 mm linearity	100 mm, 0.1 mm	Met by design
R5	Local alarm without a network	13 s	60 s	Met by design
R10	Installable by a small team	44 min estimate	45 min, hand tools	Not verifiable at TRL 3
R13	Trustworthy alarms	Needs a field record	One false warning per year or fewer	Not verifiable at TRL 3

Counts: 0 not met, 4 at risk, 4 met on paper, 2 met by design, 2 not verifiable at TRL 3, and R11 over the value-engineering target. In v0.1, R11 was not met (\$261 with a new mast); in v0.2 it was within the target (\$245) before the parts needed for construction were added.

■ Checks against the TRL 2 figures

TRL 2 claim (SLW-PRC-001 v0.2)	This note	Action
0.005 degree sensitivity	0.0055 degree output step; mode 1 needed for ± 30 degrees	Precis updated
Capsule 0.3 m deep meets R2, thin margin	Misses R2 in wet soil; 0.4 m meets it	Capsule moved to 0.4 m
Pipe conduction not included	Adds 0.05 to 0.82 K at 0.4 m	Precis updated
Airtime about 0.4 s (SF10), 1.5 s (SF12)	0.453 s and 1.810 s	Precis updated
Sensor load under 1 mW	2.67 mW with a 10 s average	Precis updated
30 min alarm about 3.2 Wh, about 2 A	3.00 Wh, 2.00 A	Precis updated
Beacon duty not set	1 % needed in precaution	Precis updated
Siren about 65 to 70 dB(A) at 100 m	65.5 to 68.5 dB(A)	Stands
Bus drop about 0.1 V	0.096 V	Stands
About 3 kB per day	363 kB for 90 days (about 4 kB per day)	Stands
About \$254 without FieldNode, \$380 with	\$261 and \$387 in v0.1; \$245 and \$371 for the reference site in v0.2 (SLW-DDR-002); \$296.50 and \$422.50 in v0.3 (SLW-DDR-003)	Precis, BOM notes and README updated
TwinKit gateway about \$285	About \$290 (TwinKit TRL 3 BOM)	Updated